

Representation Fidelity Module (RFM)

OOF™ Origin Open Foundation™

Independent Methodological Authority

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: SIMULOS® — Simulation Governance Architecture

Parent Standard: Reality Representation Standard (RRS)

Operational Layer: Simulation Reality Representation Layer

Category: Governance & Enforcement

Subcategory: Simulation Governance

Type: Parent Standard Module

Governed Space: Representation Fidelity

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AI-Readable: Yes

Authority: OOF®

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Canonical Language: English (UCL™)

Governed Space

Representation Fidelity

Canonical Definition

Representation Fidelity Module (RFM) defines the governance conditions under which the representation of selected and abstracted reality is required, assessed, bounded, and documented at a level of fidelity sufficient for the governed Simulation Purpose, Scope, scenarios, operating conditions, and intended interpretation.

RFM establishes whether the simulation representation preserves the material characteristics, behaviors, relationships, dependencies, states, and dynamics of Relevant Reality with sufficient faithfulness for the simulation inquiry without requiring unnecessary or unjustified replication of reality.

Operational Role

The preceding modules establish:

RSM — What parts of reality must the simulation represent?

RAM — How may the complexity of that selected reality legitimately be reduced?

RFM asks:

How faithfully must the resulting representation reproduce the material characteristics of the reality it claims to represent?

The governed progression is:

Reality Selection → Reality Abstraction → Fidelity Requirement → Fidelity Assessment → Fidelity Gap → Fidelity Limitation → Representation Fidelity Status → Representation Resolution

RFM does not determine whether reality was correctly selected.

It does not govern the abstraction decision itself.

It does not determine the exact representational granularity.

It governs whether the resulting representation is sufficiently faithful to the material reality required by the simulation inquiry.

Module Operational Space

RFM governs: fidelity objectives, fidelity requirements, fidelity dimensions, fidelity baselines, fidelity thresholds, physical fidelity, behavioral fidelity, structural fidelity, functional fidelity, environmental fidelity, temporal fidelity, spatial fidelity, interaction fidelity, dependency fidelity, state fidelity, process fidelity, failure fidelity, distributional fidelity, variability fidelity, boundary-condition fidelity, cross-component fidelity, scenario-relative fidelity, fidelity assessment, fidelity evidence, fidelity gaps, fidelity trade-offs, fidelity uncertainty, fidelity sensitivity, fidelity limitations, conditional fidelity, fidelity status, fidelity change, and fidelity traceability.

Module Function

A simulation can contain all relevant Reality Elements and still represent them incorrectly.

A simulation may include: the correct actor, the correct environment, the correct process, the correct dependency, and the correct interaction

while reproducing their behavior with insufficient fidelity.

RFM exists to prevent: presence being mistaken for faithful representation, visual realism being mistaken for operational fidelity, high detail being mistaken for sufficient fidelity, one fidelity dimension compensating invisibly for another, nominal behavior masking poor failure fidelity, environmental fidelity being ignored, interaction fidelity being inferred from component fidelity, fidelity requirements being selected after simulation results are known, fidelity claims exceeding the represented conditions, and simulation outputs being interpreted beyond the fidelity supported by the representation.

Fidelity Principle

SIMULOS® establishes:

Representation fidelity is sufficient only when the characteristics material to the simulation inquiry are represented faithfully enough that their remaining representation error does not invalidate the intended simulation interpretation.

Fidelity therefore does not mean:

maximum similarity to reality.

It means:

sufficient similarity in the dimensions that materially matter.

Fidelity Is Purpose-Relative

There is no universal fidelity level that is correct for every simulation.

The required fidelity depends upon:

Simulation Object + Simulation Purpose + Simulation Scope + Relevant Reality + Scenarios + Intended Interpretation

A representation sufficient for:

concept exploration

may be insufficient for:

safety-critical operational decision support.

A representation sufficient for:

aggregate traffic analysis

may be insufficient for:

autonomous collision-avoidance testing.

Fidelity Object

A Fidelity Object is the represented Reality Element, characteristic, behavior, relationship, process, state, environment, or interaction whose faithfulness to Relevant Reality requires governance.

A Fidelity Object may therefore be: an individual simulated entity, a subsystem, a behavioral process, an environmental condition, an interaction, a dependency, a failure mechanism, a population, or the combined simulation representation.

Fidelity Requirement

A Fidelity Requirement establishes how faithfully a material characteristic must be represented for the simulation inquiry.

A Fidelity Requirement should derive from:

Simulation Purpose & Scope

↓

Reality Selection

↓

Material Characteristic

↓

Approved Abstraction

↓

Required Fidelity

The requirement should not be created merely because a simulator happens to support a particular level of realism.

Fidelity Dimensions

Fidelity should not automatically be treated as one global property.

A representation may have high fidelity in one dimension and low fidelity in another.

RFM therefore permits fidelity to be governed across independent dimensions.

Physical Fidelity

Physical Fidelity concerns the degree to which relevant physical characteristics and physical relationships are represented consistently with Relevant Reality.

This may include: mass, force, acceleration, temperature, pressure, material behavior, energy transfer, friction, aerodynamics, structural response, or other physical properties.

Behavioral Fidelity

Behavioral Fidelity concerns whether simulated actors or systems behave sufficiently like their relevant real-world counterparts.

This may include: responses, decisions, transitions, adaptation, reaction patterns, control behavior, human behavior, autonomous behavior, or organizational behavior.

RFM governs representation fidelity.

It does not replace the governance of cognition, AI behavior, or autonomous

authority in their respective OOF® architectures.

Structural Fidelity

Structural Fidelity concerns whether materially relevant components and relationships are represented with sufficient correspondence to the real structure.

A structurally simplified model may remain legitimate where removed internal structure cannot materially affect the simulation inquiry.

Functional Fidelity

Functional Fidelity concerns whether the simulated object performs the materially relevant functions and produces the materially relevant responses expected from the represented reality.

A simulation may therefore possess lower structural fidelity but sufficient functional fidelity for a particular inquiry.

This distinction must remain explicit.

Environmental Fidelity

Environmental Fidelity concerns whether environmental conditions materially affecting simulation behavior are sufficiently represented.

This may include: terrain, weather, lighting, temperature, infrastructure, electromagnetic conditions, population density, traffic, resource availability, or other environmental factors.

Temporal Fidelity

Temporal Fidelity concerns whether materially relevant timing characteristics are sufficiently represented.

This may include: latency, duration, sequence, frequency, synchronization, accumulation, degradation, feedback delay, recovery time, or response time.

Spatial Fidelity

Spatial Fidelity concerns whether materially relevant: positions, distances, geometry, orientation, topology, movement, boundaries, and spatial relationships

are represented sufficiently faithfully.

Interaction Fidelity

Individual components can possess high fidelity while their interaction is represented poorly.

RFM therefore treats Interaction Fidelity as an independent governance concern.

It may include fidelity of: communication, collision, coordination, coupling, transfer, feedback, influence, negotiation, synchronization, or physical interaction.

Dependency Fidelity

Dependency Fidelity concerns whether materially relevant dependencies behave sufficiently like their real-world counterparts.

A dependency may exist in the model but still be represented incorrectly in: strength, direction, timing, conditionality, failure behavior, or propagation.

State Fidelity

State Fidelity concerns whether material operational states and transitions between them correspond sufficiently to Relevant Reality.

This becomes especially important where a system may operate in: normal, degraded, emergency, recovery, failure, maintenance, or exceptional states.

Process Fidelity

Process Fidelity concerns whether relevant sequences, workflows, transformations, or causal processes are represented with sufficient faithfulness.

Failure Fidelity

Nominal fidelity does not establish failure fidelity.

A simulation may reproduce normal operation accurately while representing: faults, degradation, cascading failure, abnormal response, emergency behavior, or recovery

poorly.

Where failure behavior is material to Simulation Purpose, RFM requires it to receive explicit fidelity treatment.

Distributional Fidelity

Where reality contains populations or stochastic behavior, fidelity may depend upon whether the represented distribution sufficiently corresponds to reality.

The average alone may be insufficient.

Material characteristics may include: variance, tails, rare events, multimodality, correlations, extreme behavior, and subgroup differences.

Variability Fidelity

Real systems may vary between: actors, units, environments, time periods, operating conditions, or repeated events.

RFM governs whether this variability must be represented and whether its

simulated form is sufficiently faithful.

Boundary-Condition Fidelity

Simulation behavior may depend strongly upon boundary conditions.

RFM therefore governs whether relevant boundary conditions sufficiently correspond to those represented by the simulation claim.

A model that performs well only within narrow boundary conditions should not receive an unrestricted fidelity representation.

Cross-Component Fidelity

Fidelity must also be evaluated across interacting simulation components.

A high-fidelity vehicle model connected to a highly simplified road, weather, sensor, or human-behavior model may produce a combined fidelity profile materially weaker than the vehicle model alone suggests.

Fidelity Baseline

A Fidelity Baseline identifies the reference against which fidelity is evaluated.

The baseline may derive from: measured reality, observed operational behavior, empirical datasets, accepted physical relationships, controlled experiments, authoritative technical data, domain models, or another sufficiently justified reference.

RFM does not govern general evidence validation.

Where the baseline itself requires formal validation, VALIDOS® provides the complementary validation governance architecture.

Fidelity Evidence

A fidelity claim should have a basis.

Possible Fidelity Evidence may include: measurement comparison, observed behavior comparison, experimental comparison, known physical relationships, historical operational data, cross-model comparison, controlled test results, or expert technical assessment.

The strength of the fidelity claim should remain bounded by the strength of the available basis.

Fidelity Threshold

A Fidelity Threshold defines the level of representational correspondence required for a particular material characteristic or fidelity dimension.

SIMULOS® does not prescribe universal thresholds.

Thresholds should derive from the simulation inquiry and consequence of

representational error.

Fidelity Assessment

A Fidelity Assessment compares the represented characteristic against its required Fidelity Requirement and applicable baseline.

The assessment asks:

Is the representation sufficiently faithful for the intended simulation inquiry?

not:

Is the representation identical to reality?

Fidelity Gap

A Fidelity Gap exists where the simulation representation differs materially from the required representation of Relevant Reality.

A gap may concern: magnitude, behavior, timing, structure, variability, interaction, dependency, state transition, environmental response, or another material characteristic.

Fidelity Gap Classification

A Fidelity Gap may be classified conceptually as:

Non-Material

Material but Acceptable

Conditional

Material and Unresolved

or:

Fidelity Undetermined

The classification must remain relative to the governed Simulation Purpose and Scope.

Fidelity Limitation

A Fidelity Limitation identifies where insufficient or uncertain fidelity restricts the legitimate interpretation or application of simulation outputs.

A limitation may restrict: scenarios, environments, populations, system states, operating ranges, failure conditions, temporal periods, or output interpretation.

Fidelity Trade-Off

Simulation design frequently involves trade-offs between fidelity and: computational cost, execution time, data availability, model complexity, interpretability, scenario scale, or repeatability.

RFM permits such trade-offs.

It does not permit them to become invisible.

Computational Constraint Rule

SIMULOS® establishes:

Computational capability may constrain achievable fidelity, but computational limitation does not redefine the fidelity required by the simulation inquiry.

If the required fidelity cannot be achieved, the appropriate result may be:

Fidelity Limitation

rather than lowering the requirement until the simulation appears sufficient.
Visual Fidelity ≠ Operational Fidelity

A simulation may appear realistic while reproducing operational behavior poorly.

Likewise, a visually simple simulation may reproduce the materially relevant dynamics with high operational fidelity.

SIMULOS® therefore establishes:

Visual realism is not evidence of sufficient simulation fidelity. Detail ≠ Fidelity

Adding detail does not necessarily improve fidelity.

Incorrect detail can reduce representational accuracy.

RFM therefore separates:

Representation Detail

from:

Representation Fidelity.

Representation Resolution is governed separately by RRM.

Fidelity Uncertainty

A Fidelity Assessment may itself contain uncertainty.

This may result from: incomplete baseline data, measurement uncertainty, changing reality, limited observations, stochastic behavior, conflicting reference evidence, or insufficient coverage.

Material fidelity uncertainty should remain explicit.

Fidelity Sensitivity

RFM may assess whether simulation outcomes are sensitive to plausible changes in fidelity.

The governing question is:

Would a plausible representation error materially alter the simulation outcome or its interpretation?

Where yes, stronger fidelity governance may be required.

Scenario-Relative Fidelity

A representation may possess sufficient fidelity for one scenario and insufficient fidelity for another.

Example:

A simplified tire model may be sufficient for:

normal-speed highway traffic analysis

but insufficient for:

emergency braking on ice.

Fidelity status should therefore remain bound to the scenarios it supports.

Conditional Fidelity

A representation may be classified as sufficiently faithful only under explicit conditions.

Possible conditions include: operating range, environment, scenario class, system configuration, population, temporal period, or simulation mode.

Representation Fidelity Status

RFM may establish a governed status such as:

Fidelity Sufficient

Fidelity Conditionally Sufficient

Fidelity Partially Sufficient

Fidelity Insufficient

or:

Fidelity Undetermined

The status should remain attached to the relevant Fidelity Object, dimension, scope, and conditions.

No Global Fidelity Inflation

A high-fidelity result for one subsystem must not automatically become a high-fidelity claim for the complete simulation.

Likewise:

High Physical Fidelity

does not automatically establish:

High Behavioral Fidelity

or:

High Environmental Fidelity.

Weakest Material Fidelity Principle

Where multiple fidelity dimensions materially affect the same simulation claim:

The simulation claim cannot assume stronger representation fidelity than its materially limiting fidelity dimension supports.

This does not require one global numerical fidelity score.

It requires material weaknesses to remain visible.

Fidelity Consistency

Fidelity assumptions and requirements across interacting simulation elements should remain mutually coherent.

A high-fidelity subsystem should not depend upon a low-fidelity representation that invalidates the inquiry without that dependency being visible.

Fidelity Change

Fidelity should be reassessed where material changes occur in: Simulation Purpose, Simulation Scope, Reality Selection, abstraction, model structure, environment, scenarios, data, operating conditions, simulation capabilities, or Relevant Reality itself.

Fidelity Change Propagation

Material fidelity changes may affect:

Representation Resolution → Representation Limitations → Assumptions → Model &

Environment → Inputs & Data → Scenario Design → Execution → Output Governance
→ Reality Correlation

Fidelity therefore forms an upstream simulation-governance dependency.

Fidelity Traceability

A governed fidelity chain should support:

Simulation Purpose → Reality Selection → Material Characteristic → Abstraction
Decision → Fidelity Requirement → Fidelity Baseline → Fidelity Assessment →
Fidelity Gap → Fidelity Status → Limitation

This makes the fidelity basis reconstructable.

Relationship to Reality Selection

RSM establishes what reality must be represented.

RFM cannot compensate for an important Reality Element that was never selected.

High fidelity of incomplete reality remains incomplete representation.

Relationship to Reality Abstraction

RAM establishes how selected reality may be simplified.

RFM evaluates whether the resulting abstraction still represents material characteristics with sufficient faithfulness.

Thus:

RAM governs permitted complexity reduction.

RFM governs the faithfulness remaining after that reduction.

Relationship to Representation Resolution

Fidelity and resolution are related but distinct.

Fidelity asks:

How faithfully does the representation correspond to Relevant Reality?

Resolution asks:

At what granularity is that representation expressed?

A high-resolution representation can have low fidelity.

A lower-resolution representation may have sufficient fidelity for the intended inquiry.

RRM governs the next space.

Relationship to Representation Limitation

RFM identifies limitations specifically arising from fidelity.

RLM subsequently governs the broader set of limitations affecting the complete Reality Representation.

Relationship to Reality Correlation

RFM establishes representation fidelity requirements before and during simulation design.

Later SIMULOS® Reality Correlation governance may compare simulation behavior or outputs with observed reality.

These are adjacent but distinct spaces.

RFM asks:

Is the representation designed and demonstrated with sufficient faithfulness?

Reality Correlation asks:

How does simulation behavior or output correspond with observed reality?

Cross-Architecture Boundary

RFM does not govern actual Operational Reality.

That remains within ORA™.

RFM does not perform general validation governance.

That remains within VALIDOS®.

RFM does not govern general integrity.

That remains within INTEGROS®.

RFM governs specifically:

the sufficiency of correspondence between simulation representation and the material characteristics of reality selected for simulation.

Minimum Implementation Framework

1. Identify Fidelity Objects

Determine which represented Reality Elements, characteristics, behaviors, interactions, dependencies, and processes require fidelity governance. 2. Establish Fidelity Requirements

Define the required correspondence for each material fidelity dimension. 3. Establish Fidelity Baselines

Identify the governed references against which fidelity can be assessed. 4.

Assess Representation Fidelity

Compare the simulation representation with the applicable requirements and baselines. 5. Identify Fidelity Gaps

Record material differences between required and achieved representation. 6. Evaluate Fidelity Impact

Determine whether identified gaps could materially affect simulation behavior, outputs, or interpretation. 7. Establish Fidelity Status

Classify fidelity as sufficient, conditional, partial, insufficient, or undetermined within the applicable scope. 8. Establish Fidelity Limitations

Bind material limitations to affected scenarios, conditions, outputs, or interpretations. 9. Assess Cross-Component Fidelity

Determine whether fidelity remains sufficient across interacting simulation elements. 10. Preserve Fidelity Traceability and Reassessment

Maintain the complete fidelity chain and reassess it after material change.

Governance Outputs

RFM may produce: Fidelity Object Register, Fidelity Requirement, Fidelity Dimension Map, Fidelity Baseline Record, Physical Fidelity Assessment, Behavioral Fidelity Assessment, Structural Fidelity Assessment, Functional Fidelity Assessment, Environmental Fidelity Assessment, Temporal Fidelity Assessment, Spatial Fidelity Assessment, Interaction Fidelity Assessment, Dependency Fidelity Assessment, State Fidelity Assessment, Process Fidelity Assessment, Failure Fidelity Assessment, Distributional Fidelity Assessment, Variability Fidelity Assessment, Boundary-Condition Fidelity Assessment, Cross-Component Fidelity Assessment, Fidelity Evidence Record, Fidelity Threshold Record, Fidelity Gap Record, Fidelity Gap Classification, Fidelity Trade-Off Record, Fidelity Uncertainty Record, Fidelity Sensitivity Assessment, Conditional Fidelity Record, Representation Fidelity Status, Fidelity Limitation Record, Fidelity Change Record, Fidelity Traceability Map, and Representation Fidelity Basis.

Use Case 1 — Autonomous Vehicle Simulation

Scenario

An autonomous-vehicle simulation represents: vehicle dynamics, sensors, pedestrians, road infrastructure, traffic, and weather.

Application

RFM separately evaluates whether the simulation possesses sufficient: physical fidelity for vehicle dynamics, behavioral fidelity for pedestrians, environmental fidelity for weather, interaction fidelity between vehicle and road users, and failure fidelity for degraded sensor conditions.

Result

The simulation may be classified:

Fidelity Sufficient — Normal Urban Operation

while remaining:

Fidelity Insufficient — Severe Sensor Degradation

The simulation cannot use its normal-operation fidelity to imply sufficient representation of failure scenarios. Use Case 2 — Spacecraft Thermal Simulation

Scenario

A spacecraft simulation is used to evaluate thermal behavior across orbital conditions. Application

RFM compares represented: solar exposure, radiation, material thermal behavior, internal heat generation, heat transfer, and environmental variation

against the fidelity required for the thermal inquiry. Result

The simulation may demonstrate sufficient thermal fidelity across normal orbital conditions while identifying a fidelity limitation for an extreme orientation not sufficiently represented by the model.

The limitation remains attached to subsequent simulation interpretation. Use Case 3 — Industrial Digital Twin

Scenario

A digital twin represents a production facility.

Individual machines are modeled accurately, but interactions between production stages are highly simplified. Application

RFM identifies:

High Component Fidelity

but:

Insufficient Interaction Fidelity

for analysis of cascading production disruption. Result

The digital twin remains suitable for individual-machine analysis but cannot legitimately claim sufficient fidelity for plant-wide cascading-failure simulation until the interaction representation is strengthened.

Architectural Position

Representation Fidelity is the third internal governance space of the Reality Representation Standard (RRS).

The complete RRS progression is:

Reality Selection → Reality Abstraction → Representation Fidelity →
Representation Resolution → Representation Limitation

RSM establishes:

What parts of reality must be represented?

RAM establishes:

How may their complexity be reduced?

RFM establishes:

How faithfully must the resulting representation correspond to the material characteristics of that reality?

RRM next establishes:

At what granularity must that representation operate?

This separation prevents selection, abstraction, fidelity, and resolution from becoming one undefined concept of simulation “realism.” Foundational Principle

A simulation representation is sufficiently faithful only when the characteristics that materially matter to the simulation inquiry remain represented closely enough to reality that the remaining fidelity gap does not invalidate the intended interpretation.

Canonical Closing Statement

Representation Fidelity Module establishes the governance conditions under which selected and abstracted reality must remain sufficiently faithfully represented for the governed Simulation Purpose and Scope. By separating fidelity from visual realism, detail, resolution, and general validation, and by governing physical, behavioral, structural, functional, environmental, temporal, spatial, interaction, dependency, state, process, failure, distributional, and variability fidelity, RFM ensures that simulation representations cannot claim greater correspondence with reality than their material fidelity dimensions, baselines, evidence, gaps, uncertainty, conditions, and limitations can actually support.

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